

Thermodynamics: Introduction to Thermodynamics, system and surrounding, basic thermodynamic terminologies (equilibrium, state, property, process, cycle), reversible and irreversible process, laws of thermodynamics (with numerical on first law of thermodynamics), comparison among heat engine and heat pump.

Introduction:

Thermodynamics is the branch of physics that deals with the relationships between heat, work, and energy. It is a fundamental discipline that has numerous applications in various fields, including engineering, physics, chemistry, and biology.

Key Concepts

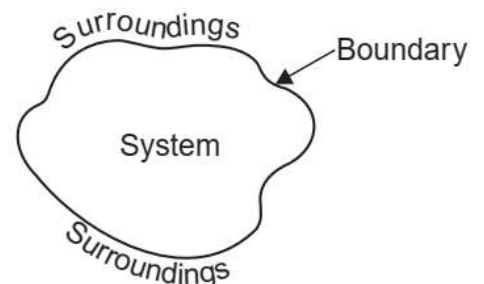
1. Energy: The ability to do work, which can exist in various forms (e.g., kinetic, potential, thermal).
2. Heat: A form of energy transfer that occurs due to a temperature difference.
3. Work: Energy transfer that occurs through a force applied over a distance.
4. Temperature: A measure of the average kinetic energy of particles in a system.

Laws of Thermodynamics

1. Zeroth Law: If two systems are in thermal equilibrium with a third system, they are also in thermal equilibrium with each other.
2. First Law: Energy cannot be created or destroyed, only converted from one form to another (conservation of energy).
3. Second Law: The total entropy of an isolated system always increases over time, and spontaneous processes occur in the direction of increasing entropy.
4. Third Law: As the temperature of a system approaches absolute zero, the entropy of the system approaches a minimum value.

System, Surrounding & Boundary

System: A system is defined as a quantity of matter or a region in space chosen for study. The mass or region outside the system is called the surroundings.



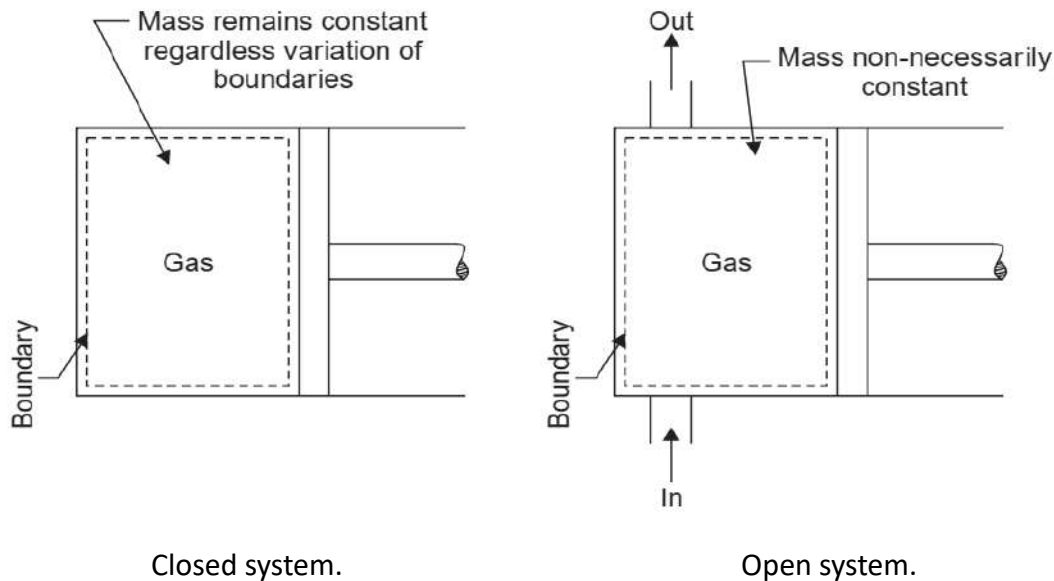
Boundary: the real or imaginary surface that separates the system from its surroundings. The boundaries of a system can be fixed or movable. Mathematically, the boundary has zero thickness, no mass, and no volume.

Closed system or control mass: consists of a fixed amount of mass, and no mass can cross its boundary. But, energy in the form of heat or work, can cross the boundary, and the volume of

a closed system does not have to be fixed. e.g. Air in rigid container, Air in piston cylinder arrangement (without valves)

Open system or control volume: is a properly selected region in space. It usually encloses a device that involves mass flow such as a compressor. Both mass and energy can cross the boundary of a control volume. e.g. Pump, Boiler, Super heater, Turbine, Condenser

Isolated system: A closed system that does not communicate with the surroundings by any means. E.g. Stationary well insulated thermo-flask



Thermodynamic Equilibrium

Two systems are said to be in thermodynamic equilibrium when they are in thermal, mechanical, chemical and phase equilibrium with each other.

1. Thermal equilibrium: The temperature of the system does not change with time and has same value at all points of the system.

2. Mechanical equilibrium. There are no unbalanced forces within the system or between the surroundings. The pressure in the system is same at all points and does not change with respect to time.

3. Chemical equilibrium. No chemical reaction takes place in the system and the chemical composition which is same throughout the system does not vary with time.

Properties of systems

A property of a system is a characteristic of the system which depends upon its state, but not upon how the state is reached. There are two sorts of property:

1. Intensive properties: These properties do not depend on the mass of the system.

Examples: Temperature, pressure, density, specific heat, specific enthalpy etc.

2. Extensive properties: These properties depend on the mass of the system.

Example: Volume, Kinetic energy, Potential energy, Internal energy (U), Enthalpy (H) and Entropy (S).

Extensive properties are often divided by mass associated with them to obtain the intensive properties.

State

State is the condition of the system at an instant of time as described or measured by its properties. Or each unique condition of a system is called a state. At a given state, all the properties of a system have fixed values. If the value of even one property changes, the state will change to a different one.

It follows from the definition of state that each property has a single value at each state. Stated differently, all properties are state or point functions. Therefore, all properties are identical for identical states.

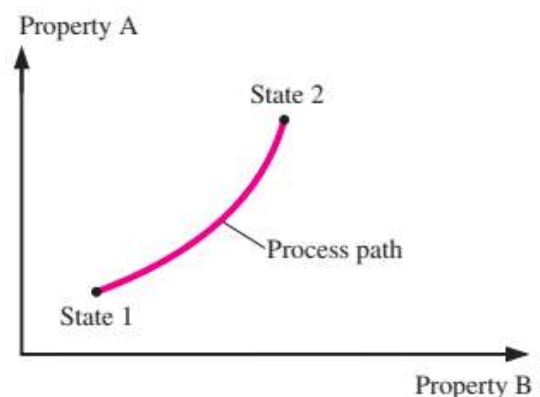
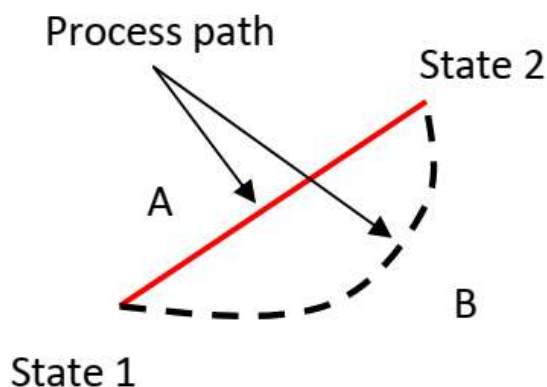
On the basis of the above discussion, we can determine if a given variable is property or not by applying the following tests:

- A variable is a property, if and only if, it has a single value at each equilibrium state.
- A variable is a property, if and only if, the change in its value between any two prescribed equilibrium states is single-valued.

Therefore, any variable whose change is fixed by the end states is a property.

Processes and Cycles

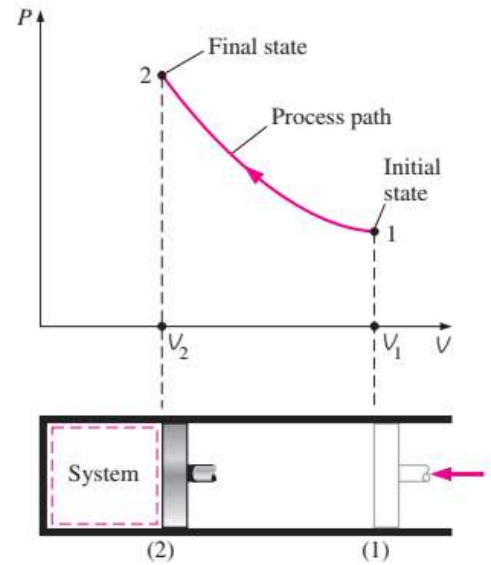
Any change a system undergoes from one equilibrium state to another is called a *process*, and the series of states through which a system passes during a process is called a *path*.



Quasistatic or Quasi-Equilibrium Process

When a process proceeds in such a manner that the system remains infinitesimally close to an equilibrium state at all times, it is called a quasistatic, or quasi-equilibrium, process. A quasi-equilibrium process can be viewed as a sufficiently slow process that allows the system to adjust itself internally so that properties in one part of the system do not change any faster than those at other parts.

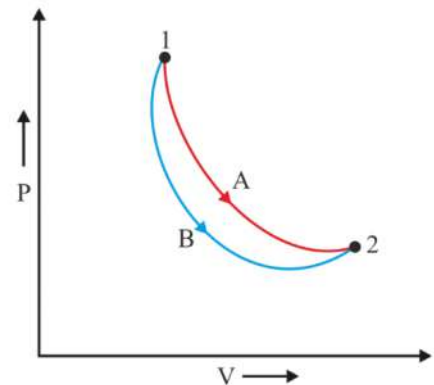
- Quasi-static process is infinitely slow process such that intensive properties are uniform throughout the system.
- All internally reversible processes are quasi-static process but all quasi-static processes need not to be internally reversible.



The P-V diagram of a compression process.

The prefix **iso-** is often used to designate a process for which a particular property remains constant. An **isothermal process**, for example, is a process during which the temperature T remains constant; an **isobaric process** is a process during which the pressure P remains constant; and an **isochoric** (or isometric) process is a process during which the specific volume v remains constant.

A system is said to have undergone a **cycle** if it returns to its initial state at the end of the process. That is, for a cycle the initial and final states are identical.



Energy, Work and Heat

Energy

Energy is a general term embracing energy in transition and stored energy. The stored energy of a substance may be in the forms of mechanical energy and internal energy (other forms of stored energy may be chemical energy and electrical energy). Part of the stored energy may take the form of either potential energy (which is the gravitational energy due to height above a chosen datum line) or kinetic energy due to velocity. The balance part of the energy is known as **internal energy**.

Heat and work are the forms of energy in transition. These are the only forms in which energy can cross the boundaries of a system. Neither heat nor work can exist as stored energy.

Work

Work is said to be done when a *force moves through a distance*.

Heat

Heat (denoted by the symbol Q), may be, defined in an analogous way to work as follows :
“Heat is ‘something’ which appears at the boundary when a system changes its state due to a difference in temperature between the system and its surroundings”.

Heat, like work, is a transient quantity which only appears at the boundary while a change is taking place within the system.

Sign conventions of heat and work

	Direction of flow	Heat (Q)	Work (W)
Supplied to the system	Into the system	Positive (+Q)	Negative (-W)
Supplied from the system	Out of the system	Negative (-Q)	Positive (+W)

Comparison of Work and Heat

Similarities:

- (i) Both are *path functions and inexact differentials*.
- (ii) Both are boundary phenomenon *i.e.*, both are recognized at the boundaries of the system as they cross them.
- (iii) Both are associated with a process, not a state. Unlike properties, work or heat has no meaning at a state.
- (iv) Systems possess energy, but not work or heat.

Dissimilarities :

- (i) In heat transfer temperature difference is required.
- (ii) In a stable system there cannot be work transfer, however, there is no restriction for the transfer of heat.
- (iii) The sole effect external to the system could be reduced to rise of a weight but in the case of a heat transfer other effects are also observed.

REVERSIBLE WORK

Let us consider an ideal frictionless fluid contained in a cylinder above a piston as shown in Fig. Assume that the pressure and temperature of the fluid are uniform and that there is no friction between the piston and the cylinder walls.

Let A = Cross-sectional area of the piston,

p = Pressure of the fluid at any instant,

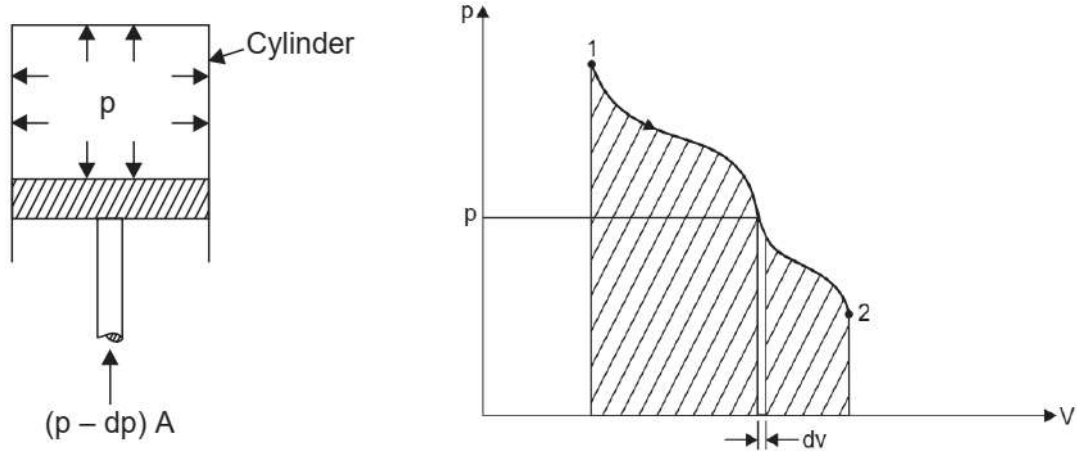
$(p - dp) A$ = Restraining force exerted by the surroundings on the piston, and

dl = The distance moved by the piston under the action of the force exerted.

Then work done by the fluid on the piston is given by force times the distance moved,

i.e., Work done by the fluid = $(pA) \times dl = pdV$

(where dV = a small increase in volume)



Or considering unit mass, Work done = $p dv$

Hence when a reversible process takes place between state 1 and state 2, we have

Work done by the unit mass of fluid = $\int_1^2 p dv$

Zeroth Law of Thermodynamics

- When a body A is in thermal equilibrium with a body B, and also separately with a body C, then bodies B and C will be in thermal equilibrium with each other. This is known as the Zeroth law of thermodynamics.
- It is the basis of temperature measurement.

Thermodynamic temperature scale, $\frac{T_1 - I_1}{S_1 - I_1} = \frac{T_2 - I_2}{S_2 - I_2}$

Where, T_1 = Temperature on first scale (in $^{\circ}\text{C}$)

T_2 = Temperature on second scale (in $^{\circ}\text{C}$)

S_1 = Steam point temperature on first scale

S_2 = Steam point temperature on second scale

I_1 = Ice point temperature on first scale

I_2 = Ice point temperature on first scale

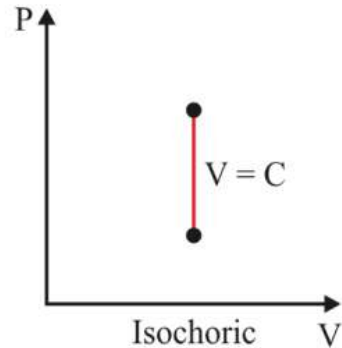
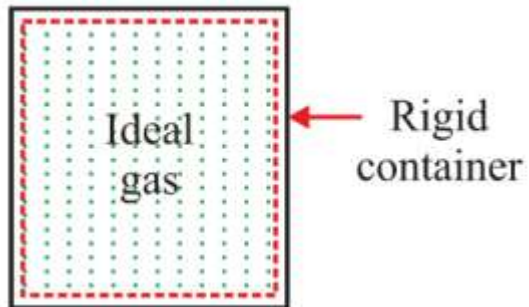
e.g.
$$\frac{^{\circ}\text{C}}{5} = \frac{^{\circ}\text{F} - 32}{9} = \frac{K - 273.15}{5}$$

Various thermodynamic processes

Constant Volume Process / Isochoric Process

$V = \text{Constant}$ [Any Substance]

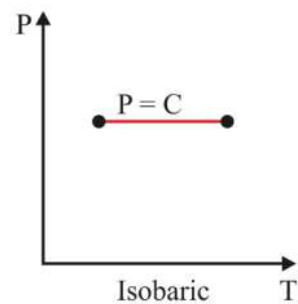
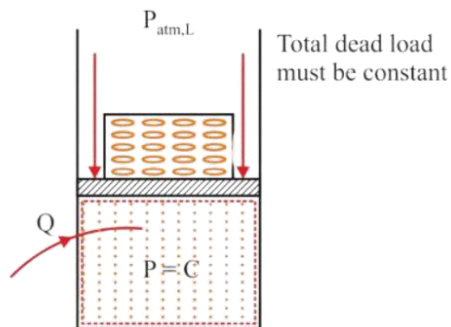
$dV = 0$ [Any Substance]



Constant pressure process/ Isobaric Process

The pressure is constant at equilibrium states.

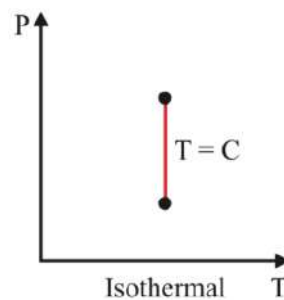
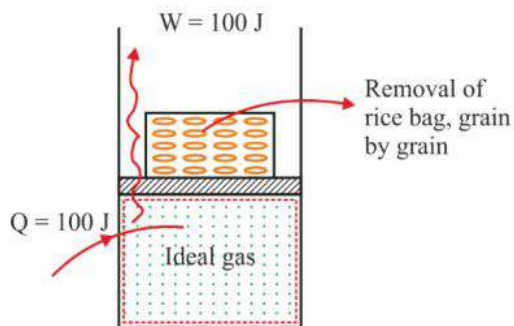
- $P = \text{constant}$
- $dP = 0$ [Any substance]

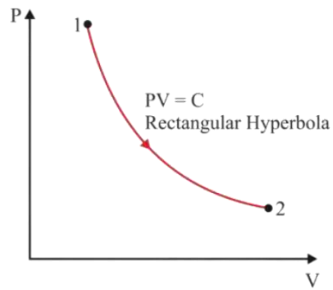


Constant temperature process/Isothermal Process

$T = \text{Constant}$ [Any Substance]

$dT = 0$ [Any Substance]





Adiabatic process

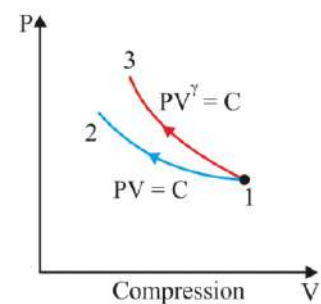
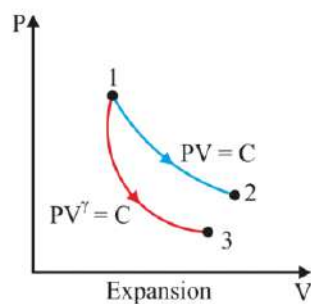
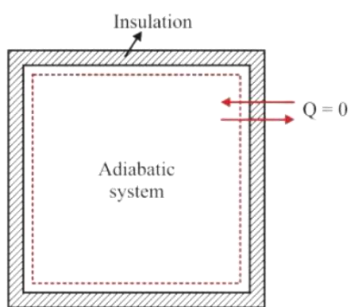
A process during which there is no heat interaction is called an adiabatic process.

- $Q = 0$ (Any Substance)
- For perfect gas undergoing Isentropic process following relations can be used

$$PV^\gamma = \text{Constant}$$

$$PV^{(1-\gamma)/\gamma} = \text{Constant}$$

$$TV^{(\gamma-1)} = \text{Constant}$$



Polytropic Process

For Polytropic process following equations can be used

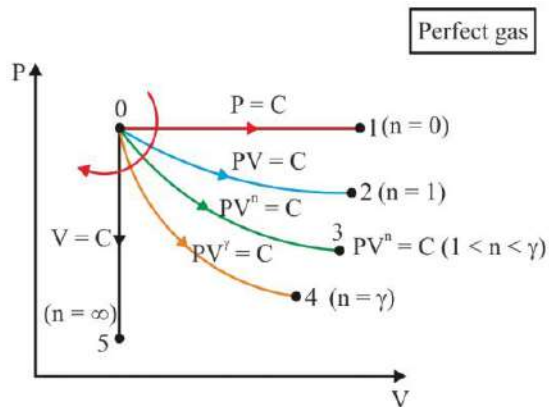
$$PV^n = \text{Constant} \quad \text{----- Any substance}$$

$$PV^{(1-n)/n} = \text{Constant} \quad \text{----- Ideal Gas only}$$

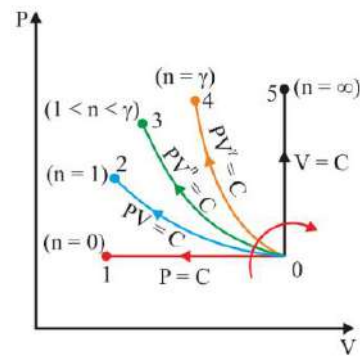
$$TV^{(n-1)} = \text{Constant} \quad \text{----- Ideal Gas only}$$

Where n is known as polytropic index

Representation of various processes on same P-V diagram



P-V diagram for expansion process



P-V diagram for compression process

Name	Characteristic	Index (n of $PV^n = c$)
Isobaric	Pressure (P) = const.	$n = 0$
Isothermal	Temperature (T) = const.	$n = 1$
Adiabatic	Heat (H/Q) = const.	$n = \gamma$ (cp / cv)
Polytropic	----	$n = n$
Isochoric	Volume (V) = const.	$n = \infty$

First Law of Thermodynamics

- First law of thermodynamics is the law of conservation of energy. It gives the concept of internal energy.

Internal Energy

It is the heat energy stored in a gas. Joule's law of internal energy states that internal energy of a perfect gas is a function of temperature only. In other words, internal energy of a gas is dependent on the temperature change only and is not affected by the change in pressure and volume.

We do not know how to find the absolute quantity of internal energy in any substance; however, what is needed in engineering is the change of internal energy (ΔU).

First Law of Thermodynamics for a closed system undergoing a cycle

For a closed system undergoing a cyclic process, net heat interaction is equal to the net work interaction.

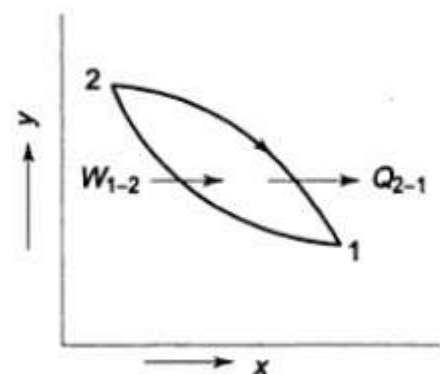
For a closed system undergoing cyclic process

$$\Sigma Q = \Sigma W \quad (\text{Any cycle})$$

or

$$\oint \delta Q = \oint \delta W \quad (\text{Any cycle})$$

The First Law of Thermodynamics may also be stated as follows:



“Heat and work are mutually convertible but since energy can neither be created nor destroyed, the total energy associated with an energy conversion remains constant”.

Or

— “No machine can produce energy without corresponding expenditure of energy, *i.e.*, it is impossible to construct a perpetual motion machine of first kind”.

Consequences of First Law of Thermodynamics

- (1) Heat and work interactions are energies in transit which can only be observed at the boundaries. Both are path functions.
- (2) Energy of the system is the property of the system.
- (3) Energy of an isolated system is constant.
- (4) Perpetual motion machine of first kind (PMM-1) is impossible.

First Law of Thermodynamics for a closed system undergoing a process

The expression, $\Sigma Q = \Sigma W$ applies only to systems undergoing cycles, and the algebraic sum of all the energy transfer across system boundaries is zero. But if a system undergoes a change of state during which heat and work transfer are involved, the net energy transfer will be stored or accumulated within the system. If Q is the amount of heat transferred to the system and W is the amount of work transferred from the system during the process, the net energy ($Q - W$) will be stored in the system. Energy in storage is neither heat nor work, and is given the name internal energy or simply the energy of the system.

Therefore, $Q - W = \Delta E$ or $Q = W + \Delta E$

Where ΔE is the increase in the energy.

In the differential form, the above equation can be written as, $\delta Q - \delta W = dE$

But for a simple compressible, stationary system undergoing internally reversible process

$$\delta Q = PdV + dU$$

on integrating, $Q = \int (PdV) + \Delta U$

Drawback of First Law of Thermodynamics

First law of thermodynamics gives the information about the conservation of energy during a process, but it doesn't give any information about the feasibility & nature of the process. The first law of thermodynamics sets no limit to the amount of the total energy of a system which can be caused to flow out as work.

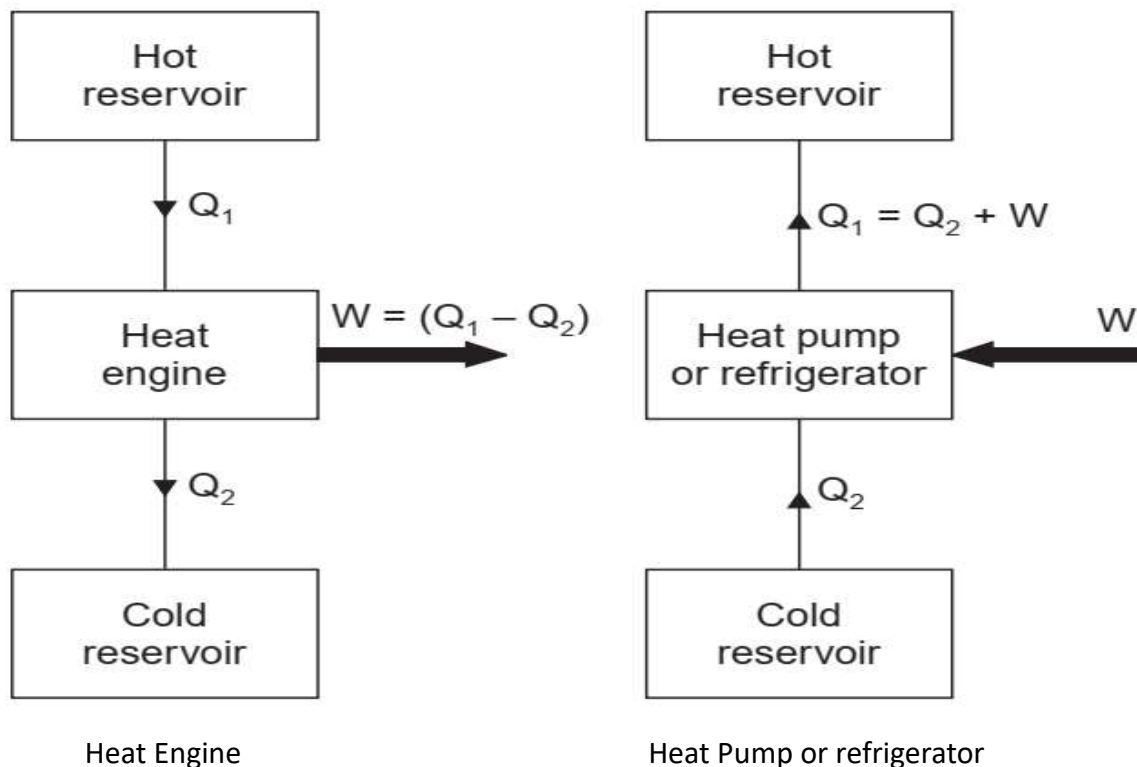
Further the first law of thermodynamics establishes equivalence between the quantity of heat used and the mechanical work but does not specify the conditions under which conversion of heat into work is possible, neither the direction in which heat transfer can take place. This gap has been bridged by the second law of thermodynamics.

Performance of heat engines, heat pump and refrigerator

A *heat engine* is used to produce the maximum work transfer from a given positive heat transfer. The measure of success is called the *thermal efficiency* of the engine and is defined by the ratio:

$$\text{Thermal efficiency, } \eta_{th} = \frac{W}{Q_1}$$

where, W = Net work transfer from the engine, and
 Q_1 = Heat transfer to engine



For a *reversed heat engine* acting as a **refrigerator** when the purpose is to achieve the maximum heat transfer from the cold reservoir, the measure of success is called the *co-efficient of performance* (C.O.P.). It is defined by the ratio:

$$\text{Co-efficient of performance, (C.O.P.)}_{ref.} = Q_2 / W$$

For a reversed heat engine acting as a **heat pump**, the measure of success is again called the co-efficient of performance. It is defined by the ratio:

$$\text{Co-efficient of performance, (C.O.P.)}_{HP} = Q_1 / W$$

where,

Q_1 = Heat transfer to hot reservoir,

Q_2 = Heat transfer from cold reservoir and

W = Net work transfer to the heat pump / refrigerator.

In all the above three cases application of the first law gives the relation $Q_1 - Q_2 = W$, and this can be used to rewrite the expressions for thermal efficiency and co-efficient of performance solely in terms of the heat transfers.

$$\eta_{th} = \frac{Q_1 - Q_2}{Q_1}$$

$$COP_{ref} = \frac{Q_2}{Q_1 - Q_2}$$

$$COP_{HP} = \frac{Q_1}{Q_1 - Q_2}$$

From the above relations, it may also be concluded that,

$$\eta_{th} = 1/COP_{HP}$$

$$COP_{HP} = 1 + COP_{ref}$$

Reversible Processes

A reversible process should fulfill the following *conditions*:

1. The process should not involve friction of any kind.
2. Heat transfer should not take place with finite temperature difference.
3. The energy transfer as heat and work during the forward process should be identically equal to energy transfer as heat and work during the reversal of the process.
4. There should be no free or unrestricted expansion.
5. There should be no mixing of the fluids.
6. The process must proceed in a series of equilibrium states.

Some examples of ideal reversible processes are:

- i. Frictionless adiabatic expansion or compression;
- ii. Frictionless isothermal expansion or compression;

iii. Condensation and boiling of liquids.

Some examples of irreversible processes are:

- i. Combustion process;
- ii. Mixing of two fluids;
- iii. All processes involving friction;
- iv. Flow of electric current through a resistance;
- v. Heat flow from a higher temperature to lower temperature.

Reversible processes are preferred because the devices which produce work such as engines and turbines, reversible process of the working fluid delivers more work than the corresponding irreversible processes. Also, in case of fans, compressors, refrigerators and pumps less power input is required when reversible processes are used in place of corresponding irreversible ones.

In thermodynamic analysis concept of reversibility, though hypothetical, is very important because a reversible process is the most efficient process. Only reversible processes can be truly represented on property diagrams. Thermodynamic reversibility can only be approached but can never be achieved. Thus, the main task of the engineer is to design the system which will evolve approximate reversible processes.

Statements of second law of thermodynamics

Clausius Statement

"It is impossible for a self-acting machine working in a cyclic process unaided by any external agency, to convey heat from a body at a lower temperature to a body at a higher temperature".

In other words, heat of, itself, cannot flow from a colder to a hotter body.

Kelvin-Planck Statement

"It is impossible to construct an engine, which while operating in a cycle produces no other effect except to extract heat from a single reservoir and do equivalent amount of work". Although the Clausius and Kelvin-Planck statements appear to be different, they are really equivalent in the sense that a *violation of either statement implies violation of other.*

Numerical problems

Q1. The properties of a closed system change following the relation between pressure and volume as $pV = 3.0$ where p is in bar V is in m^3 . Calculate the work done when the pressure increases from 1.5 bar to 7.5 bar.

Solution. Initial pressure, $p_1 = 1.5$ bar

Final pressure, $p_2 = 7.5$ bar

Relation between p and V , $pV = 3.0$

Work done, W :

The work done during the process is given by

$$W = \int_{V_1}^{V_2} p dV$$

$$V_1 = \frac{3.0}{p_1} = \frac{3.0}{1.5} = 2 \text{ m}^3$$

$$V_2 = \frac{3.0}{p_2} = \frac{3.0}{7.5} = 0.4 \text{ m}^3$$

$$\begin{aligned} \therefore W &= 10^5 \int_2^{0.4} \frac{3.0}{V} dV \text{ Nm} & [\because 1 \text{ bar} = 10^5 \text{ N/m}^2] \\ &= 10^5 \times 3.0 [\log_e V]_2^{0.4} = 10^5 \times 3.0 (\log_e 0.4 - \log_e 2) \\ &= -3 \times 10^5 \log_e (2/0.4) = -3 \times 10^5 \times \log_e 5 = -3 \times 10^5 \times 1.61 \text{ Nm} \\ &= -4.83 \times 10^5 \text{ Nm} = -4.83 \times 10^5 \text{ J} = -483 \text{ kJ. (Ans.)} \end{aligned}$$

Q2. To a closed system 150 kJ of work is supplied. If the initial volume is 0.6 m³ and pressure of the system changes as $p = 8 - 4V$, where p is in bar and V is in m³, determine the final volume and pressure of the system.

Solution. Amount of work supplied to a closed system = 150 kJ

Initial volume = 0.6 m³

Pressure-volume relationship, $p = 8 - 4V$

The work done during the process is given by

$$\begin{aligned} W &= \int_{V_1}^{V_2} p dV \\ &= 10^5 \int_{0.6}^{V_2} (8 - 4V) dV = 10^5 \left[8V - 4 \times \frac{V^2}{2} \right]_{0.6}^{V_2} \\ &= 10^5 [8(V_2 - 0.6) - 2(V_2^2 - 0.6^2)] \\ &= 10^5 [8V_2 - 4.8 - 2V_2^2 + 0.72] \\ &= 10^5 [8V_2 - 2V_2^2 - 4.08] \text{ Nm or J} \end{aligned}$$

But this work is equal to -150×10^3 J as this work is supplied to the system.

$$\therefore -150 \times 10^3 = 10^5 [8V_2 - 2V_2^2 - 4.08]$$

$$\text{or } 2V_2^2 - 8V_2 + 2.58 = 0$$

$$V_2 = \frac{8 \pm \sqrt{64 - 4 \times 2 \times 2.58}}{4} = \frac{8 \pm 6.585}{4} = 0.354 \text{ m}^3$$

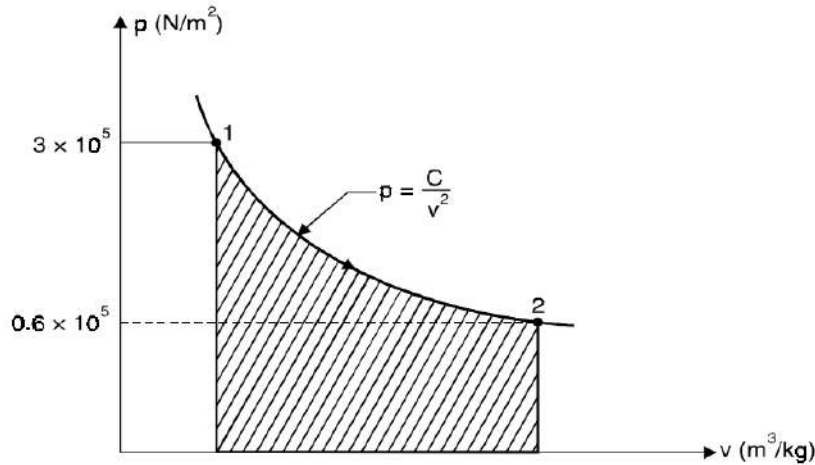
Positive sign is incompatible with the present problem, therefore it is not considered.

$$\therefore \text{Final volume, } V_2 = 0.354 \text{ m}^3. \text{ (Ans.)}$$

$$\text{and, final pressure, } p_2 = 8 - 4V = 8 - 4 \times 0.354$$

$$= 6.584 \text{ bar} = 6.584 \times 10^5 \text{ N/m}^2 \text{ or Pa. (Ans.)}$$

Q3. A fluid at a pressure of 3 bar, and with specific volume of 0.18 m³/kg, contained in a cylinder behind a piston expands reversibly to a pressure of 0.6 bar according to a law, $p = C/v^2$ where C is a constant. Calculate the work done by the fluid on the piston.



$$p_1 = 3 \text{ bar} = 3 \times 10^5 \text{ N/m}^2$$

$$v_1 = 0.18 \text{ m}^3/\text{kg}$$

$$\text{Work done} = \text{Shaded area} = \int_1^2 p \, dv$$

$$\text{i.e.,} \quad \text{Work done, } W = \int_1^2 \frac{C}{v^2} \, dv = C \int_1^2 \frac{dv}{v^2} = C \left[\frac{v^{-2+1}}{-2+1} \right]_{v_1}^{v_2}$$

$$= C \left[-v^{-1} \right]_{v_1}^{v_2} = C \left[-\frac{1}{v} \right]_{v_1}^{v_2} = C \left[\frac{1}{v_1} - \frac{1}{v_2} \right]$$

Also

$$C = pv^2 = p_1 v_1^2 = 3 \times 0.18^2 = 0.0972 \text{ bar (m}^3/\text{kg)}^2$$

and

$$v_2 = \sqrt{\frac{C}{p_2}} = \sqrt{\frac{0.0972}{0.6}} = 0.402 \text{ m}^3/\text{kg}$$

Substituting the values of C , v_1 and v_2 in eqn. (i), we get

$$\begin{aligned} \text{Work done, } W &= 0.0972 \times 10^5 \left[\frac{1}{0.18} - \frac{1}{0.402} \right] \text{ Nm/kg} \\ &= 29840 \text{ Nm/kg. (Ans.)} \end{aligned}$$

Q4. In an internal combustion engine, during the compression stroke the heat rejected to the cooling water is 50 kJ/kg and the work input is 100 kJ/kg. Calculate the change in internal energy of the working fluid stating whether it is a gain or loss.

Solution. Heat rejected to the cooling water, $Q = -50 \text{ kJ/kg}$

(-ve sign since heat is rejected)

Work input, $W = -100 \text{ kJ/kg}$

(-ve sign since work is supplied to the system)

Using the relation, $Q = (u_2 - u_1) + W$

$$-50 = (u_2 - u_1) - 100$$

or

$$u_2 - u_1 = -50 + 100 = 50 \text{ kJ/kg}$$

Hence, **gain in internal energy = 50 kJ/kg. (Ans.)**

Q5. 0.3 kg of nitrogen gas at 100 kPa and 40°C is contained in a cylinder. The piston is moved compressing nitrogen until the pressure becomes 1 MPa and temperature becomes 160°C. The work done during the process is 30 kJ. Calculate the heat transferred from the nitrogen to the surroundings. C_v for nitrogen = 0.75 kJ/kg K.

Solution. Mass of nitrogen, $m = 0.3 \text{ kg}$

Temperature before compression = 40°C or 313 K

Temperature after compression = 160°C or 433 K

The work done during the compression process, $W = -30 \text{ kJ}$

According to first law of thermodynamics,

$$\begin{aligned} Q &= \Delta U + W = (U_2 - U_1) + W \\ &= mc_v (T_2 - T_1) + W \\ &= 0.3 \times 0.75(433 - 313) - 30 = -3 \text{ kJ} \end{aligned}$$

Hence, **heat 'rejected' during the process = 3 kJ. (Ans.)**

Note. Work, W has been taken -ve because it has been supplied from outside.

Q6. Air enters a compressor at 10^5 Pa and 25°C having volume of $1.8 \text{ m}^3/\text{kg}$ and is compressed to $5 \times 10^5 \text{ Pa}$ isothermally. Determine:

- (i) Work done;
- (ii) Change in internal energy; and
- (iii) Heat transferred.

Solution. Initial pressure of air, $p_1 = 10^5 \text{ Pa}$
 Initial temperature of air, $T_1 = 25 + 273 = 298 \text{ K}$
 Final pressure of air, $p_2 = 5 \times 10^5 \text{ Pa}$
 Final temperature of air, $T_2 = T_1 = 298 \text{ K}$ (isothermal process)

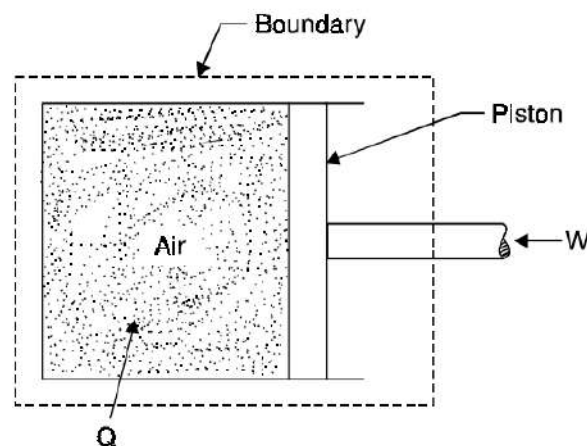
Since, it is a closed steady state process, we can write down the first law of thermodynamics

as,

$$Q = (u_2 - u_1) + W \text{per kg}$$

(i) For isothermal process :

$$W_{1-2} = \int_1^2 p \cdot dv = p_1 v_1 \log_e \left(\frac{p_1}{p_2} \right)$$



as

$$p_1 v_1 = p_2 v_2 \text{ for isothermal process}$$

$$\begin{aligned} \therefore W_{1-2} &= -10^5 \times 1.8 \log_e \left(\frac{1 \times 10^5}{5 \times 10^5} \right) \\ &= -2.897 \times 10^5 = -289.7 \text{ kJ/kg.} \end{aligned}$$

(- ve sign indicates that the work is supplied to the air)

$\therefore$ Work done on the air = 289.7 kJ/kg. (Ans.)

(ii) Since temperature is constant,

$$\therefore u_2 - u_1 = 0$$

$\therefore$ Change in internal energy = zero. (Ans.)

$$\begin{aligned} \text{(iii) Again, } Q_{1-2} &= (u_2 - u_1) + W \\ &= 0 + (-289.7) = -289.7 \text{ kJ} \end{aligned}$$

(- ve sign indicates that heat is lost from the system to the surroundings)

$\therefore$ Heat rejected = 289.7 kJ/kg. (Ans.)

Q6. A cylinder containing the air comprises the system. Cycle is completed as follows:

(i) 82000 N-m of work is done by the piston on the air during compression stroke and 45 kJ of heat are rejected to the surroundings.

(ii) During expansion stroke 100000 N-m of work is done by the air on the piston.

Calculate the quantity of heat added to the system.

Solution.

Compression stroke. Process 1-2 :

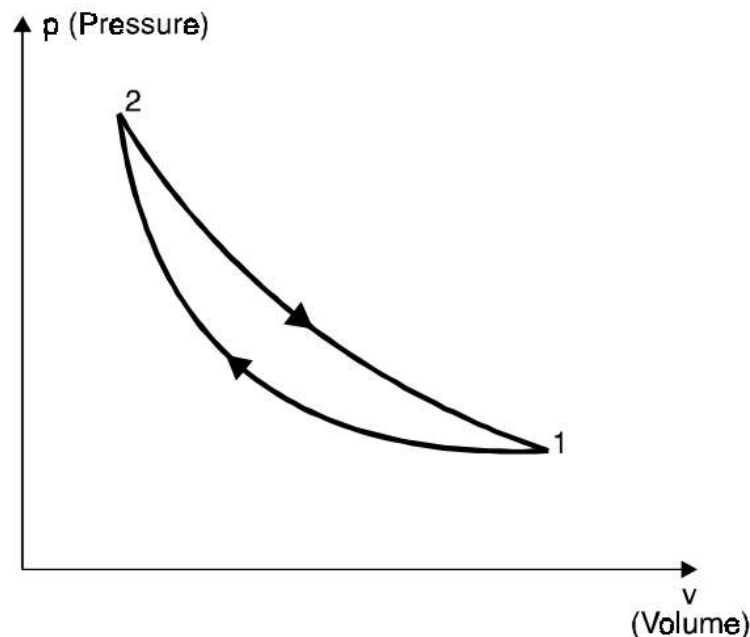
Work done by the piston *on the air*, $W_{1-2} = -82000 \text{ N-m} (= -82 \text{ kJ})$

Heat rejected to the system, $Q_{1-2} = -45 \text{ kJ}$

$$\begin{aligned} \text{Now, } Q_{1-2} &= (U_2 - U_1) + W \\ -45 &= (U_2 - U_1) + (-82) \end{aligned}$$

$$\therefore (U_2 - U_1) = 37 \text{ kJ} \quad \dots(i)$$

Expansion stroke. Process 2-1 :



Work done by air on the piston, $W_{2-1} = 100000 \text{ N-m} (= 100 \text{ kJ})$

$$\begin{aligned} \text{Now, } Q_{2-1} &= (U_1 - U_2) + W \\ &= -37 + 100 \text{ kJ} = 63 \text{ kJ} \end{aligned}$$

Hence, **quantity of heat added to the system = 63 kJ. (Ans.)**

Q7. When a system is taken from state l to state m, in Fig. 4.18, along path lqm, 168 kJ of heat flows into the system, and the system does 64 kJ of work:

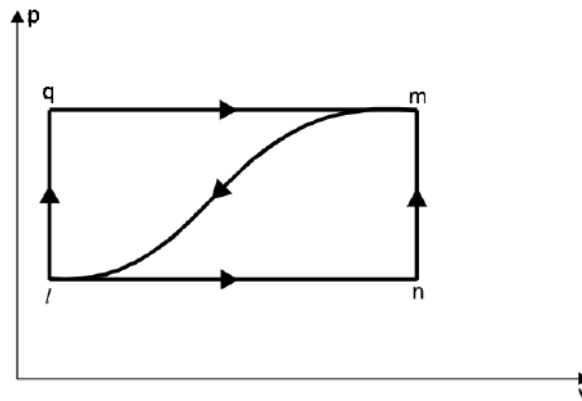
(i) How much will be the heat that flows into the system along path lnm if the work done is 21 kJ?

(ii) When the system is returned from m to l along the curved path, the work done on the system is 42 kJ. Does the system absorb or liberate heat, and how much of the heat is absorbed or liberated?

(iii) If $U_l = 0$ and $U_n = 84 \text{ kJ}$, find the heat absorbed in the processes ln and nm.

Solution.

$$\begin{aligned} Q_{l-q-m} &= 168 \text{ kJ} \\ W_{l-q-m} &= 64 \text{ kJ} \end{aligned}$$



We have,

$$\begin{aligned} Q_{l-q-m} &= (U_m - U_l) + W_{l-q-m} \\ 168 &= (U_m - U_l) + 64 \end{aligned}$$

$$\therefore U_m - U_l = 104 \text{ kJ. (Ans.)}$$

$$\begin{aligned} \text{(i) } Q_{l-n-m} &= (U_m - U_l) + W_{l-n-m} \\ &= 104 + 21 = 125 \text{ kJ. (Ans.)} \end{aligned}$$

$$\begin{aligned} \text{(ii) } Q_{m-l} &= (U_l - U_m) + W_{m-l} \\ &= -104 + (-42) = -146 \text{ kJ. (Ans.)} \end{aligned}$$

The system liberates 146 kJ.

$$\begin{aligned} \text{(iii) } W_{l-n-m} &= W_{l-n} + W_{n-m} = W_{l-m} = 21 \text{ kJ} \\ &[\because W_{n-m} = 0, \text{ since volume does not change.}] \end{aligned}$$

$$\begin{aligned} \therefore Q_{l-n} &= (U_n - U_l) + W_{l-n} \\ &= (84 - 0) + 21 = 105 \text{ kJ. (Ans.)} \end{aligned}$$

$$\text{Now } Q_{l-m-n} = 125 \text{ kJ} = Q_{l-n} + Q_{n-m}$$

$$\begin{aligned} \therefore Q_{n-m} &= 125 - Q_{l-n} \\ &= 125 - 105 = 20 \text{ kJ. (Ans.)} \end{aligned}$$

Exercise Questions:

1. When the valve of the evacuated bottle is opened, atmospheric air rushes into it. If the atmospheric pressure is 101.325 kPa, and 1.2 m³ of air (measured at atmospheric conditions) enters into the bottle, calculate the work done by the air. [Ans. – 60.8 kJ]
2. A pump forces 1.2 m³/min of water horizontally from an open well to a closed tank where the pressure is 0.9 MPa. Compute the work the pump must do upon the water in an hour just to force the water into the tank against the pressure. [Ans. 15972 kJ]
3. 1 kg of a fluid is compressed reversibly according to a law $pv = 0.25$ where p is in bar and v is in m³/kg. The final volume is 1/4 of the initial volume. Calculate the work done on the fluid and sketch the process on a p - v diagram. [Ans. 34660 Nm]
4. A gas having a volume of 0.05 m³ and pressure of 6.9 bar expands reversibly in a cylinder behind a piston according to law $pv^{1.2} = \text{constant}$ until the volume is 0.08 m³. Calculate the work done by the gas. Also sketch the process on a p - v diagram. [Ans. 15300 Nm]
5. In a cyclic process, heat transfers are + 14.7 kJ, – 25.2 kJ, – 3.56 kJ and + 31.5 kJ. What is the net work for this cyclic process? [Ans. 17.34 kJ]
6. A domestic refrigerator is loaded with food and the door closed. During a certain period, the machine consumes 1 kWh of energy and the internal energy of the system drops by 5000 kJ. Find the net heat transferred in the system. [Ans. – 8.6 MJ]
7. A closed system of constant volume experiences a temperature rise of 20°C when a certain process occurs. The heat transferred in the process is 18 kJ. The specific heat at constant volume for the pure substance comprising the system is 1.2 kJ/kg°C, and the system contains 2 kg of this substance. Determine the change in internal energy and the work done. [Ans. $\Delta U = 48$ kJ; $W = -30$ kJ]
8. A stationary mass of gas is compressed without friction from an initial state of 2 m³ and 2×10^5 N/m² to a final state of 1 m³ and 2×10^5 N/m², the pressure remaining the same. There is a transfer of 360 kJ of heat from the gas during the process. How much does the internal energy of the gas change? [Ans. $\Delta U = -160$ kJ]
9. A piston cylinder arrangement has a gas in the cylinder space. During a constant pressure expansion to a larger volume the work effect for the gas is 1.6 kJ, the heat added to the gas and cylinder arrangement is 3.2 kJ and the friction between the piston and cylinder wall amounts to 0.24 kJ. Determine the change in internal energy of the entire apparatus (gas, cylinder, piston). [Ans. 1.84 kJ]
10. A system receives 42 kJ of heat while expanding with volume change of 0.123 m³ against an atmosphere of 12 N/cm². A mass of 80 kg in the surroundings is also lifted through a distance of 6 metres.
 - (i) Find the change in energy of the system.
 - (ii) The system is returned to its initial volume by an adiabatic process which requires 100 kJ of work. Find the change in energy of system.
 - (iii) Determine the total change in energy of the system.

[Ans. (i) 22.54 kJ, (ii) 100 kJ, (iii) 122.54 kJ]

Unit-3

Refrigeration & Air- conditioning

Fundamentals of Mechanical Engineering
(BME 101 / BME 201)

Prepared by:-

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ABESEC

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- 1) Introduction to refrigeration
- 2) Methods of refrigeration
- 3) Unit of refrigeration
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- 7) Simple vapour compression refrigeration system (VCRS)
- 8) Air - Conditioning :- Introduction, its meaning and application
- 9) Psychrometry terminologies.
- 10) Human comfort conditions
- 11) Domestic refrigerator
- 12) Window air - conditioner
- 13) Numerical problems

1)

①

Refrigeration Introduction:-

Refrigeration may be defined as the process of achieving & maintaining a temperature below the surrounding / atmospheric conditions.

The purpose of the same is usually preservation of food items, maintaining human comfort conditions, maintaining a particular temperature or ~~at~~ atmospheric condition required for a specific purpose something like agriculture, industrial application etc.

2) Methods of refrigeration:- History of refrigeration is centuries old, and over the period of this various methods have been evolved to achieve the same. Some of those are as follows,

* Natural refrigeration systems.

(i) Nocturnal cooling

(ii) Evaporative cooling

(iii) Cooling by salt solution

* Artificial refrigeration systems

(i) Vapour compression refrigeration system
(VCRS)

(ii) Vapour absorption refrigeration system
(VARS)

(iii) Solar energy based refrigeration system

(iv) Gas cycle refrigeration system

(v) Steam jet refrigeration system

(vi) Thermoelectric refrigeration system

(vii) Vortex tube refrigeration system

3) Unit of refrigeration (Ton of refrigeration)

Generally the refrigerating capacity of a system is represented in terms of ton of refrigeration (TR).

It's defined as, the amount of heat that needs to be extracted from 2000 pounds of water at 0°C to convert it into ice at 0°C in 24 hours.

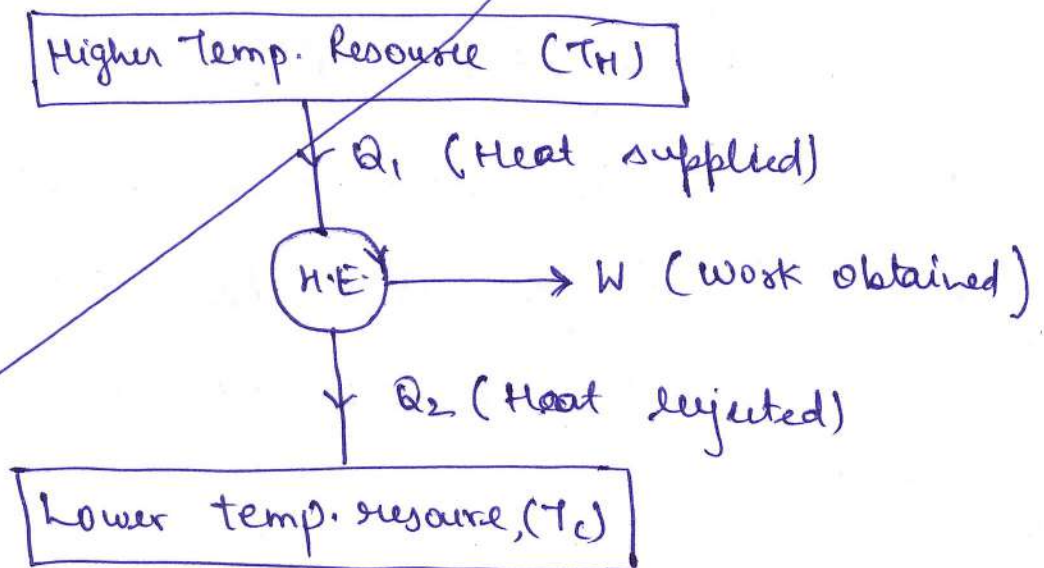
$$\begin{aligned}
 \text{So, } 1 \text{ TR} &= m L \\
 &= (2000 \times 0.453) \times 335 / 24 \times 60 \\
 &= 210 \text{ KJ/min} \\
 &= 3.5 \text{ KJ/sec } \textcircled{\text{or}} \text{ KW}
 \end{aligned}$$

where, $L = 335 \text{ KJ/kg}$, the latent heat of ice at 0°C

1TR may also be defined as, the amount of heat required to melt the 2000 pound ice at 0°C into water.

5) Comparison among Heat engine, Refrigerator and Heat pump:-

4.1 → Heat engine:-



4) Coefficient of performance (COP):-

Coefficient of performance (COP) is the ratio of desired effect to the work supplied to the system.

For a refrigeration system, desired effect could be the amount of heat extracted from the lower temperature body,

$$\boxed{COP = \frac{Q}{W}}$$

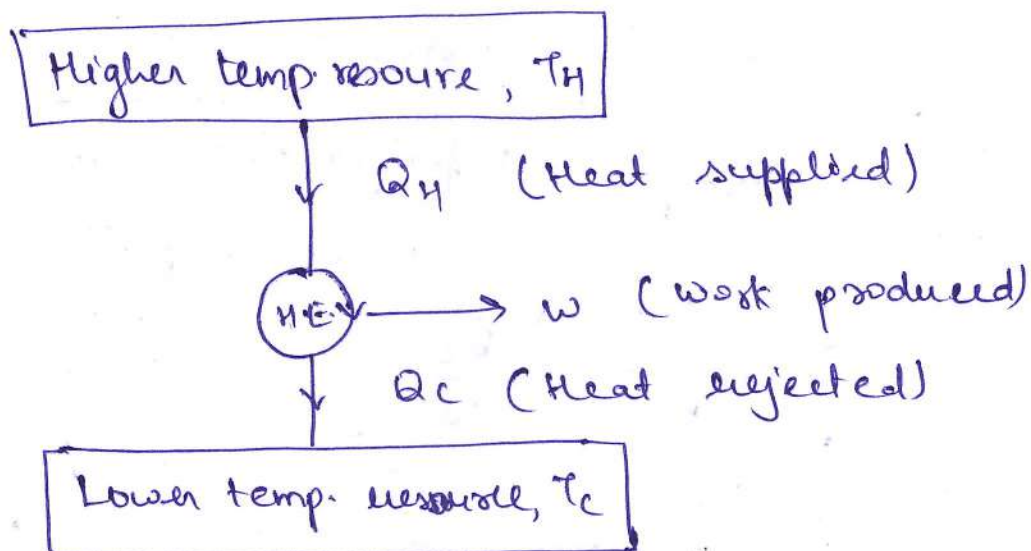
Where,

Q = heat extracted (refrigerating effect or refrigerating capacity)

W = work supplied to the system

5) Comparison among heat engine, refrigerator and heat pump:-

*) Heat engine:-

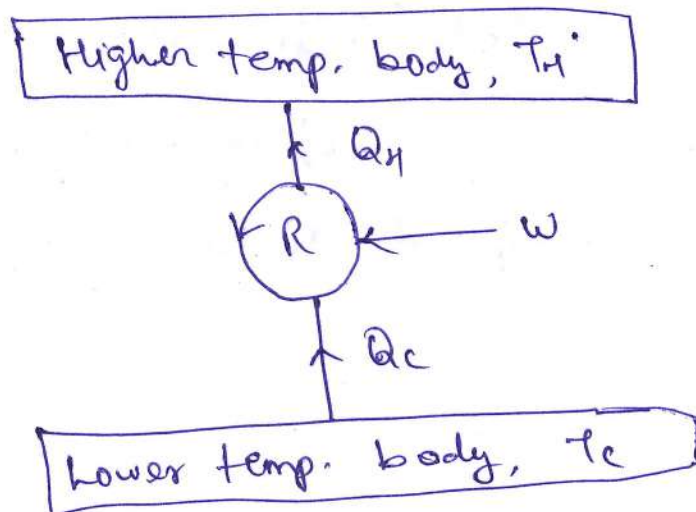


Heat engine is a thermodynamic device, which convert energy in heat form to some useful work. It's performance can be shown in terms of efficiency, thus.

$$\eta_{HE} = \frac{\text{Work output}}{\text{Heat supplied}} = \frac{W}{Q_H} = \frac{Q_H - Q_c}{Q_H} \quad \text{--- (I)}$$

$$\boxed{\eta_{HE} = 1 - \frac{Q_c}{Q_H}}$$

* Refrigerator:-



Refrigerator is the device which absorb heat (refrigerating effect) from lower temp. body to cool it down and reject the same to higher temp. body by consuming external work, thus

$$(\text{COP})_R = \frac{\text{Refrigerating effect}}{\text{Work supplied}}$$

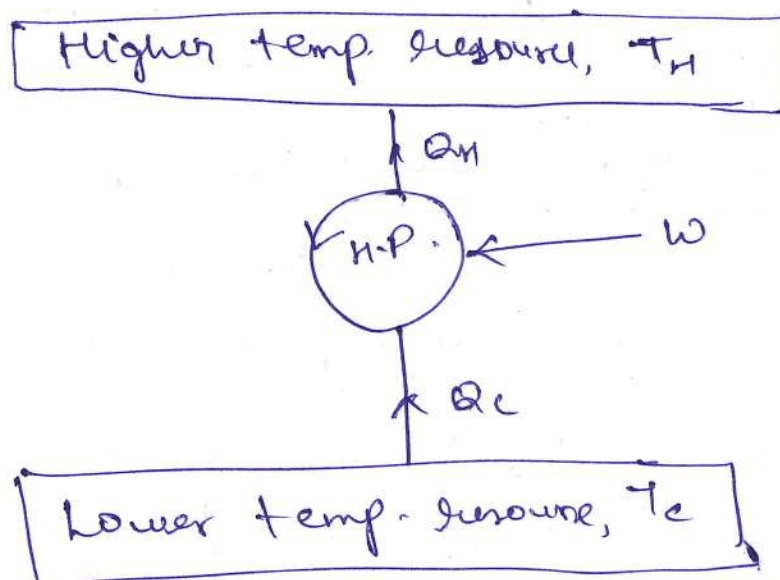
$$(\text{COP})_R = \frac{Q_C}{W}$$

$$\text{or } (\text{COP})_R = \frac{Q_C}{Q_H - Q_C}$$

— (2)

* Heat pump →

(4)



Thermodynamically, heat pump is similarly to refrigerator. i.e. absorbing heat from a cooler region and rejecting it to a higher temp. region by consuming external work.

But the desired effect of a heat pump is to heat up the space (higher temp. region) rather than cooling the lower temp region as in case of ~~refrig~~ refrigerator.

$$\text{So, } \boxed{(\text{COP})_{HP} = \frac{Q_H}{W} = \frac{Q_H}{Q_H - Q_c}} \quad \text{--- (2)}$$

Thus, a heat pump is different from a refrigerator in two aspects.

(i) Desired output — For a refrigeration the desired output/effect is to cool down

the space (lower temp. region) where as a heat pump is used for heating purpose (higher temp. region)

(ii) Working temp. ranges:- A refrigerator works between at atmospheric temp. & sub-atmospheric temp; which a heat pump works between atmospheric temp and above atmospheric condition.

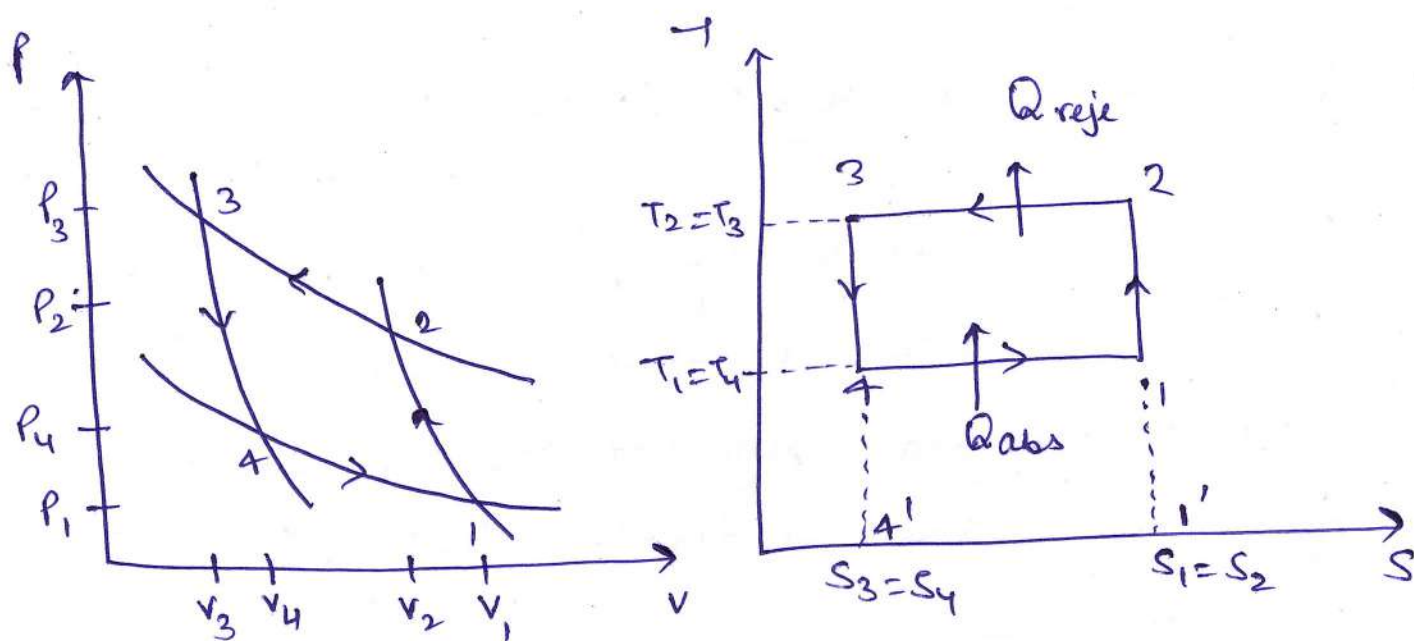
from eqⁿ (1), (2) & (3),

$$\begin{aligned} (COP)_{HP} &= 1 + (COP)_R \\ (COP)_{HP} &= 1/\eta_{HE} \end{aligned}$$

— (4)

(6) Air refrigeration system based on reverse Carnot cycle:-

A reverse Carnot cycle based on air as a medium is shown in the following figure.



Process 1-2 $\rightarrow$ Isentropic Compression

Process 2-3 $\rightarrow$ Isothermal Heat rejection Condenser

Process 3-4 $\rightarrow$ Isentropic Expansion

Process 4-1 $\rightarrow$ Isothermal heat absorption

*) Isentropic Compression process (1→2):-

In this process the air ~~is~~ is compressed isentropically from state 1 to 2, resulting in an increase in pressure and temperature while a decrease in volume. During the process the heat interaction remains zero.

*) Isothermal heat rejection (2→3) During this process the compressed air, ~~comp~~ coming from compressor, rejects its heat to the surrounding. The amount of heat rejected can be calculated from the area under the curve (2-3) of T-s diagram

$$Q_r = Q_{2-3} = \text{area } 1'234'$$

$$\boxed{Q_r = T_2(s_2 - s_3)} \quad \text{--- (1)}$$

*) Isentropic Expansion process (3→4):- Now the air is expanded so that its pressure and temperature can be reduced to evaporator level. As a result of which P & T are reduced while volume increases.

*) Isothermal heat absorption process (4-1):-

Now the air is expanded further in this process. The amount of heat absorbed,

$$Q_{\text{abs}} = Q_{4-1} = \text{area } 4'1'1'1'$$

$$\boxed{Q_{\text{ab}} = T_1 (S_1 - S_4)} \quad - (2)$$

As we know that,

Work supplied = heat rejected - heat supplied

$$W_{\text{in}} = Q_R - Q_{\text{ab}}$$

$$= T_2 (S_2 - S_3) - T_1 (S_1 - S_4)$$

$$W_{\text{in}} = T_2 (S_1 - S_4) - T_1 (S_1 - S_4), \text{ as, } S_1 = S_2 \text{ and } S_3 = S_4$$

$$\boxed{W_{\text{in}} = (T_2 - T_1) (S_1 - S_4)} \quad - (3)$$

Now, (COP) of the refrigerator

$$(COP)_R = \frac{\text{heat absorbed}}{\text{Work supplied}}$$

$$(COP)_R = \frac{T_1 (S_1 - S_4)}{(T_2 - T_1) (S_1 - S_4)}$$

$$\boxed{(COP)_R = \frac{T_1}{T_2 - T_1}} \quad - (4)$$

Eqⁿ(4) represents the (COP) of a refrigerator working on reverse Carnot cycle.

Although Carnot cycle shows the maximum COP between any two temperature ranges, it is not possible to build a refrigerating machine based on this cycle. This is due to the fact that one of the two processes is isentropic, which needs to be extremely fast, and other one is isothermal which needs to be extremely slow. This variation in speed b/w extreme ranges is impracticable to achieve in a single machine.

(7)

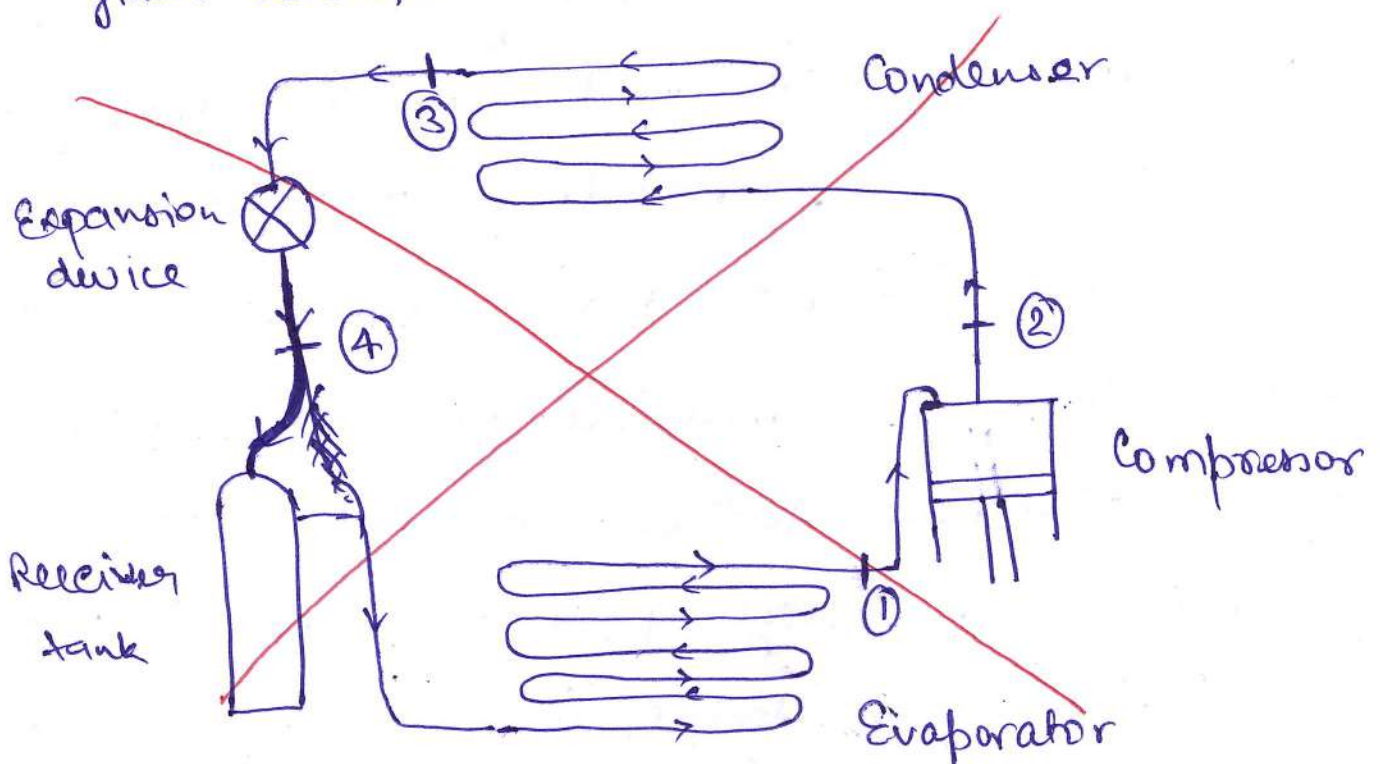
(7) Simple Vapour Compression Refrigeration System (VCRS):-

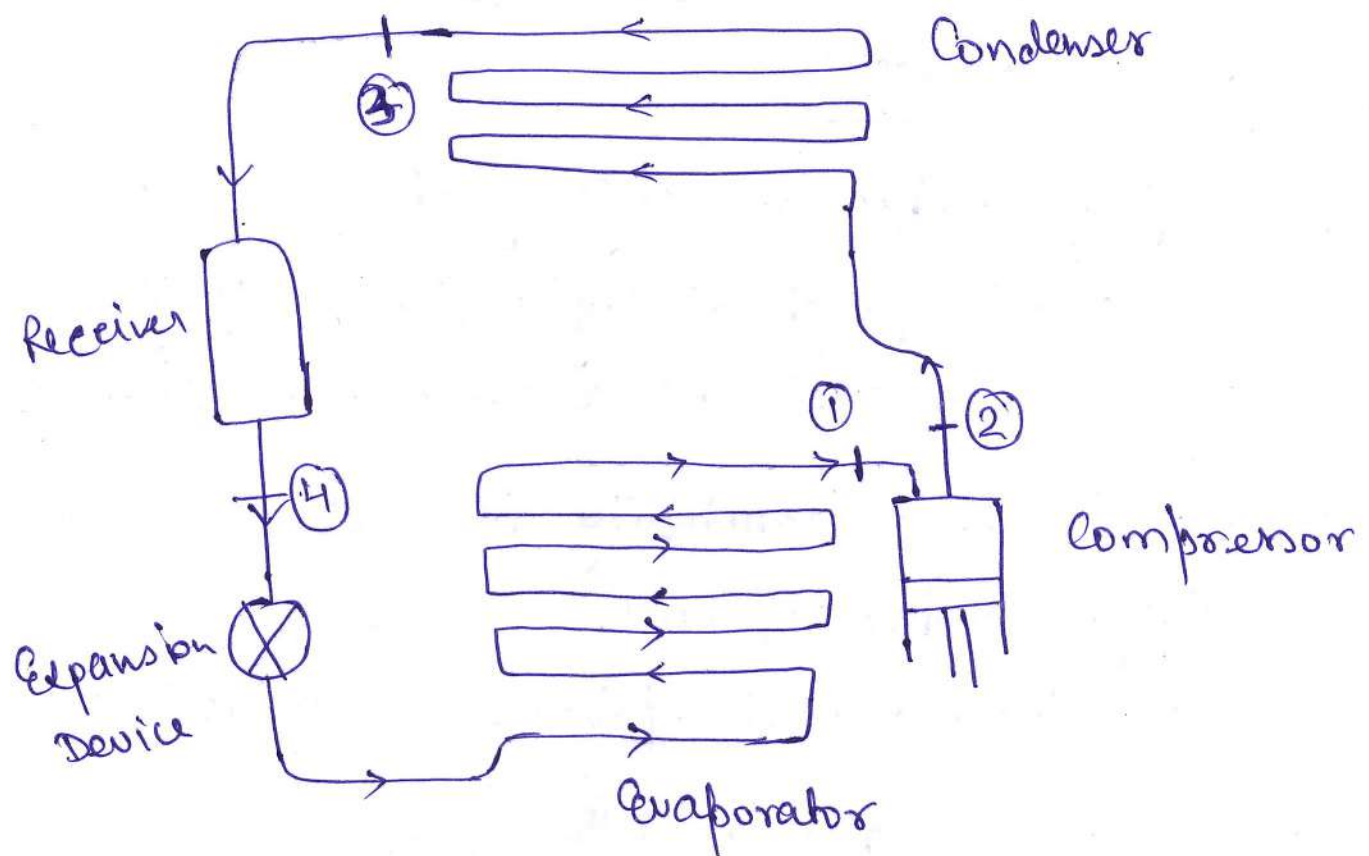
Vapour compression refrigeration cycle is the most practical cycle for making a cooling device. This is because the liquids have enormous capability to absorb heat as they evaporate/boil.

The fundamental processes required in a VCR system are as follows:-

- i) Evaporation ; (ii) Compression
- iii) Condensation ; (iv) Expansion

The flow diagram of the same is given below,-





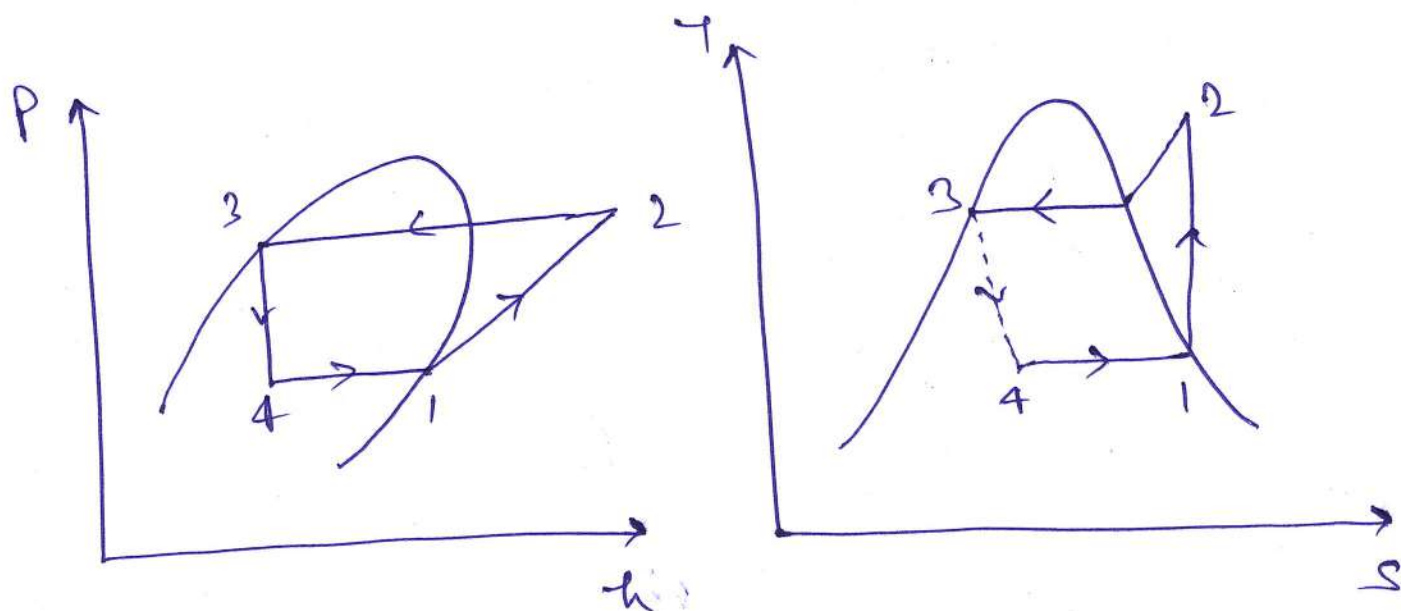
Low temperature & low pr.

- Refrigerant vapours coming out of evaporator (state 1) enters the compressor where it gets compressed isentropically. As a result of which the pr. & temp. becomes considerably higher than the surrounding conditions.

This vapour (state 2) now enters the condenser where it rejects sensible as well as latent of condensation to the surroundings and gets converted into high pressure low temperature liquid. This gets stored in the receiver tank.

The pressure of this liquid coming out of condenser (state 3) is lowered to evaporator level by using an expansion device (usually a throttling) (state 4). Finally this low pr., low temp. liquid refrigerant absorbs a ^{latent} heat of evaporation ~~to~~ from the objects to be cooled down.

Thermodynamic cycle of VCRS :-



Process 1-2 $\Rightarrow$ Isentropic Compression

Process 2-3 $\Rightarrow$ Isobaric Condensation (heat rejection)

Process 3-4 $\Rightarrow$ Isenthalpic Expansion

Process 4-1 $\Rightarrow$ Isobaric Evaporation (heat absorption)

Refrigerating effect, $q_1 = (h_1 - h_4)$

work supplied, $w = (h_2 - h_1)$

$$\therefore, \boxed{(\text{COP})_R = \frac{q_1}{w} = \left(\frac{h_1 - h_4}{h_2 - h_1} \right)}$$

(8) Air-conditioning:- Introduction, its meaning & applications

Unlike the refrigeration, the purpose of air conditioning is not just to low the temperature of the air. But to control the temperature, humidity, purity and velocity of the air in the designated space.

(9) Psychrometry:- The air surrounding us is believed to be a mixture of dry air and water vapour. i.e. the moist air. Thus the understanding of this moist air becomes of paramount importance.

The study of the same is called Psychrometrics.

Definitions:-

(i) Dry air - it is a mechanical mixture of gases present in the environment.

composition of the same is as follows:

Sr. No.	Constituent gas	By Volume (approx)	By weight (approx)	molecular mass
1.	Nitrogen (N_2)	78%.	79%.	28
2.	Oxygen (O_2)	23%.	21%.	32
3.	Argon (Ar)	0.09%.	1.3%.	40
4.	Carbon Di-oxide (CO_2)	0.03%.	0.05%.	44
5.	Hydrogen (H_2)	0.01%.	-	2

On the basis of above data, the molecular mass of dry air is assumed to be 28.97 and gas constant as 0.287 kJ/kg K .

Dry air is never really found in ~~at~~ general. So, the term air usually means 'moist air'.

Water vapour in air exists ~~to~~ in super-heated state, and is unsaturated in nature.

(ii) Saturated air: - The air is said to be saturated when it has diffused maximum possible water vapour for a given temperature and pressure.

(iii) Degree of saturation: - It's the ratio of mass of water vapour present in one kg of dry air to the mass of water vapour in the same mass when it is saturated at the same temperature and pressure.

It's also defined as the ratio of actual ^{sp.} humidity and the specific humidity of a saturated air.

(iv) Specific humidity or humidity ratio: -

It is defined as the mass of water vapour present in unit mass of dry air

sp. humidity,
$$\omega = \frac{m_w}{m_a}$$

in terms of pressure;

$$W = 0.622 \left(\frac{P_v}{P_t - P_v} \right)$$

where, $m_v \rightarrow$ mass of water vapour

$m_a \rightarrow$ mass of dry air

$P_v \rightarrow$ partial pressure of water vapour

$P_t \rightarrow$ Total atmospheric pressure.

(V) Relative humidity (R.H., ϕ) :- It's the ratio of partial pressure of water vapour (P_v) to the saturation pressure (P_s) of pure water at the same temperature & pressure.

$$\phi = \frac{P_v}{P_s}$$

It may also be defined as

$\phi = \frac{\text{mass of water vapour in a given volume of dry air}}{\text{mass of water vapour in the same volume when it is saturated at the same temp.}}$

** RH (ϕ) of a ~~can~~ saturated air is 100%.

(vi) Dry bulb temperature (t_{dbt}) :- It's the temperature of the air recorded by a thermometer when its bulb is kept dry.

(vii) Wet bulb temperature (t_{wbt}) :- WBT is the temperature of the air recorded by thermometer when its bulb is covered by a wet cloth ~~and~~ cotton wool.

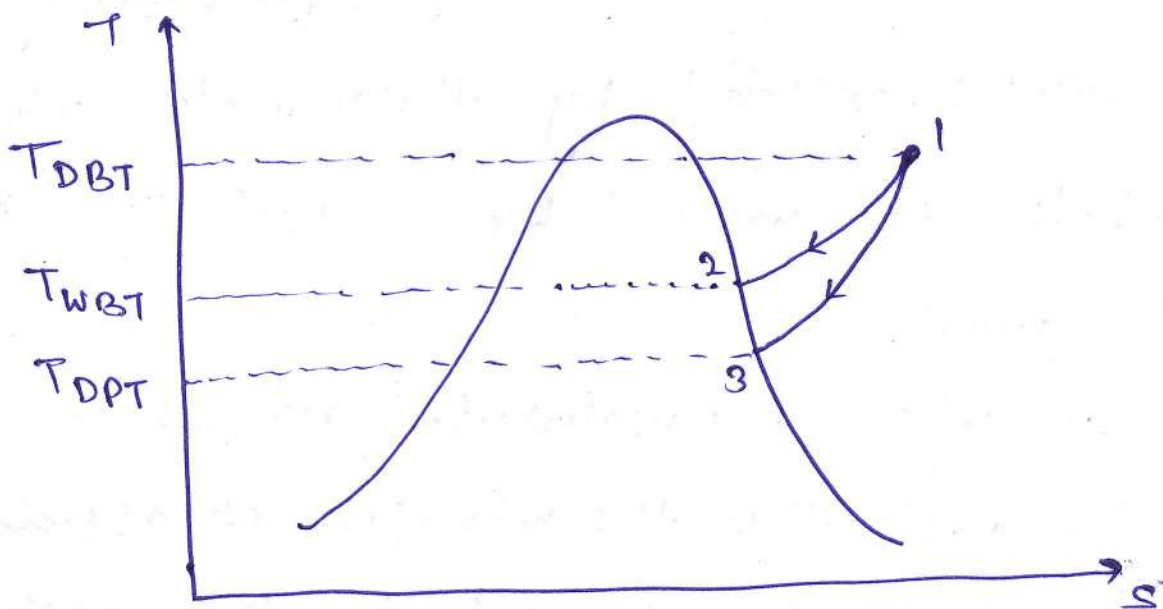
So, when a unsaturated air is made to flow past this thermometer, ~~it absorbs~~ ~~latent~~ ~~heat~~ water present in the wick absorbs latent heat ~~of~~ ~~from~~ from the air and evaporates. Thus the temperature of the air decreases while the humidity increases.

This temperature can be ~~achieved~~ achieved by cooling the unsaturated air adiabatically.

till the saturation occurs.

(viii) Dew point temperature (DPT) $\rightarrow$

It is the temperature at which the water vapour present in the air ~~starts~~ starts to condense when cooled at constant pressure.



1-2 $\rightarrow$ adiabatic saturation process

1-3 $\rightarrow$ ~~constant~~ isobaric saturation process

10) Human Comfort conditions:-

As per ASHRAE (American Society of Heating, Refrigerating and Air Conditioning Engineers)

thermal comfort conditions are, "the conditions of mind that expresses satisfaction with thermal environment."

The various factor that affect the comfort conditions, and must be taken care of while designing summer and/or winter AC are as follows

- (i) Effective temperature (DBT, sp. humidity, Air velocity)
- (ii) Metabolic heat rate
- (iii) Heat & moisture loss from the body
- (iv) Quality & quantity of air
- (v) Age
- (vi) Gender
- (vii) Type of activity
- (viii) Type of clothing

(ix) Health condition

(x) Surface temperature which are in contact

(xi) Heat received by convection & Radiation

etc.

In a usual case, following conditions are acceptable for air conditioning

- *) Effective temperature: 20°C to 26°C
- *) A dew point temperature of 2°C to 17°C
- *) Humidity 30% to 70%.
- *) Air velocity of upto 0.25 m/s

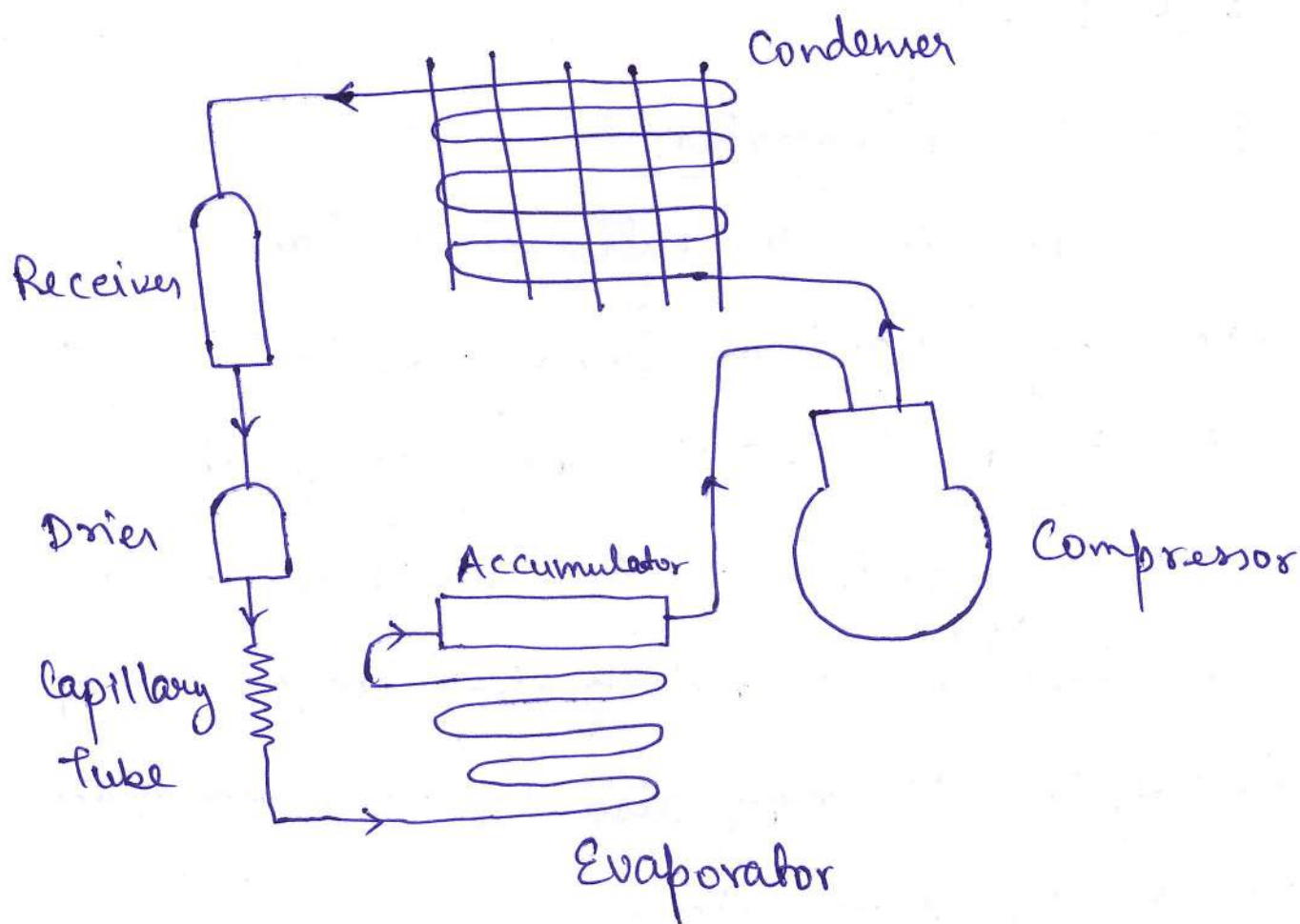
§ Construction

(11) § Domestic Refrigerator:-

At present, the refrigerators have become a necessity than a luxury product. It enables us to preserve food. The primary function of the refrigerator is to maintain a low temperature in the designated space to keep the edibles. It may also be used to make ice & keep frozen items.

Usually domestic refrigerators work on vapour compression refrigeration cycle. Now a days R134a ($C_2H_2F_4$) or R32 (CH_2F_2) are the two most widely use refrigerants.

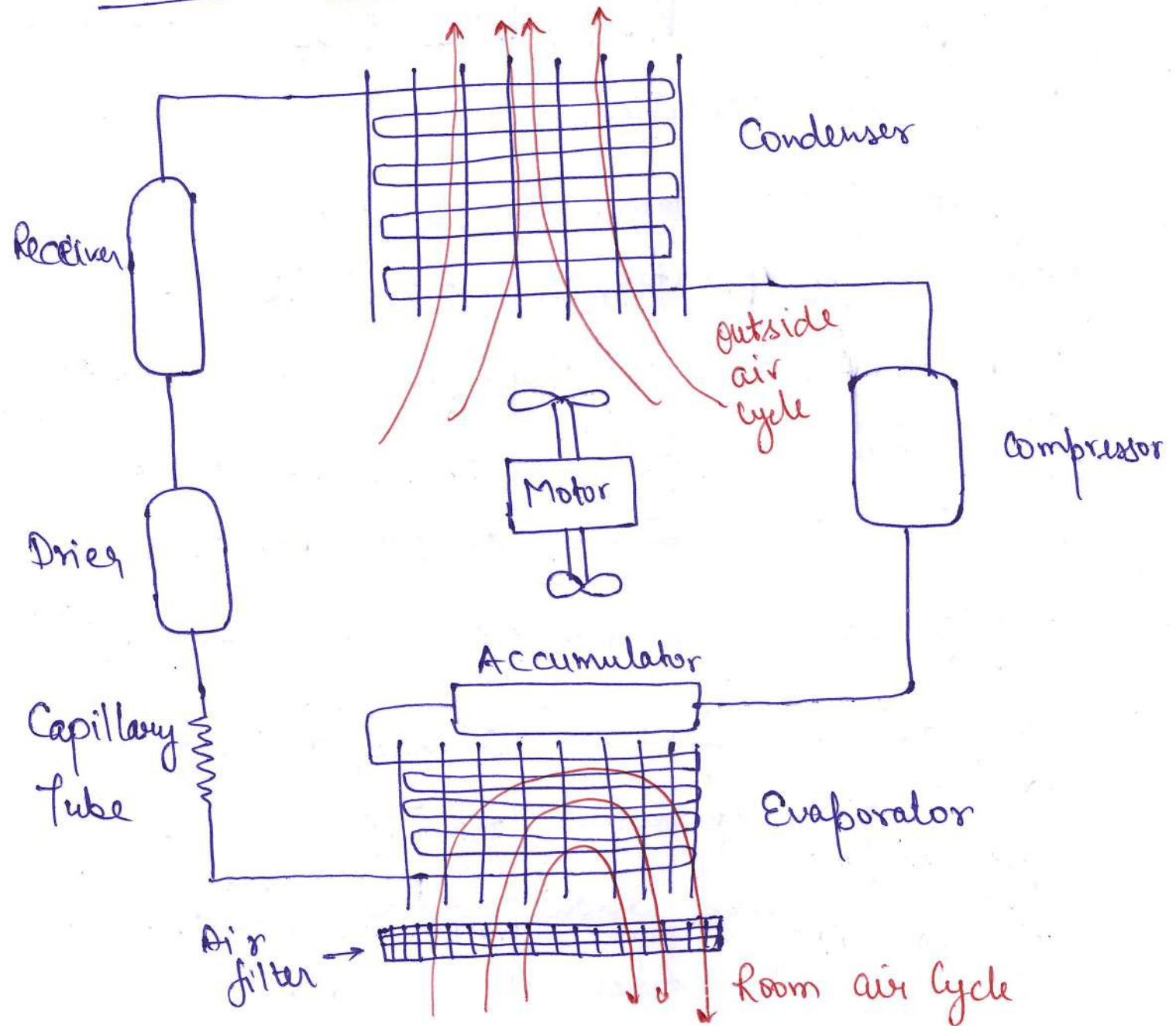
The evaporator coils are wound around the freezer section of the refrigerator. As the refrigerant flows through the evaporator, it absorbs heat from the objects and evaporate by consuming latent heat of evaporation. As liquids are high incompressible,



it needs to be filtered out before the refrigerant enters the compressor. This function is achieved by the accumulator. The dry vapours of refrigerant coming out of accumulator are then compressed to a much higher level than the surrounding conditions so that superheated vapours of refrigerant can easily reject heat to surrounding by air cooled condenser. Receiver & drier work as the storage tank and filter of refrigerant respectively. Finally, the capillary tube works as the expansion device.

which reduces the pressure & temperature of the refrigerant to the pre-evaporator level. Thus the refrigerant can be used for further cooling cycle.

(12) § Window air conditioner:-



A window air conditioner is a self contained air-conditioning unit which is installed in a window or wall opening. All the necessary devices i.e. evaporator, compressor, Receiver, condenser, electric controls etc., are fitted into the same box.

The evaporator and condenser coils are usually made up of copper ~~with~~ with aluminium as the fins attached to it.

At the room air is ~~withdrew~~ withdrawn into the evaporator, due to fan/blower behind evaporator coils, it passes through the air filter first, which removes dust particles from the air. Since the temperature of refrigerant is much lesser than the air, it absorbs sensible as well as latent heat of evaporation from the air. Thus the air cools down to the required values. Also, since the temperature of

15
of evaporator coils is below the dew point temperature, the water vapours present in the room air also gets condensed. Thus a cooler and drier air is obtained which is pushed back into the room to provide desired cooling effect.

The surrounding air is also pushed to flow over the condenser coils, so that the rate of heat ~~tran~~ transfer from condenser coils to surrounding can be increased by means of convection.

Numerical problems! -

Q1 → Find the COP of a refrigeration system if the work input is 60 kJ/kg and the refrigerating effect produced is 130 kJ/kg .

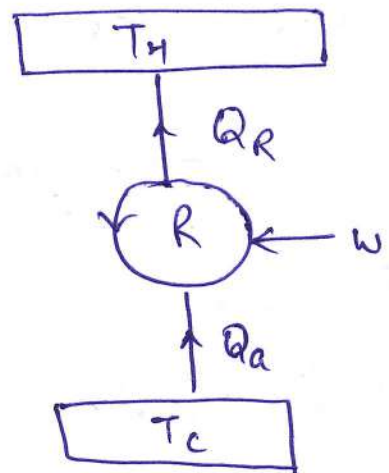
Solution:- Given data-

Refrigerating Effect, $Q_a = 130 \text{ kJ/kg}$

Work input, $w = 60 \text{ kJ/kg}$

as we know,

$$\begin{aligned} (\text{COP})_R &= \frac{Q_a}{w} \\ &= \frac{130}{60} = 2.17 \end{aligned}$$



Q2 → A refrigeration working on reverse Carnot cycle has a C.O.P. of 4. If it works as a heat pump and consumes 1 kW . Calculate the heating effect produced by the pump.

Solution →

$$(\text{COP})_R = 4$$

Power consumed, $w = 1 \text{ kW}$

(IES-2003)

as we know,

$$\begin{aligned} (\text{COP})_{\text{HP}} &= 1 + (\text{COP})_R \\ &= 1 + 4 \end{aligned}$$

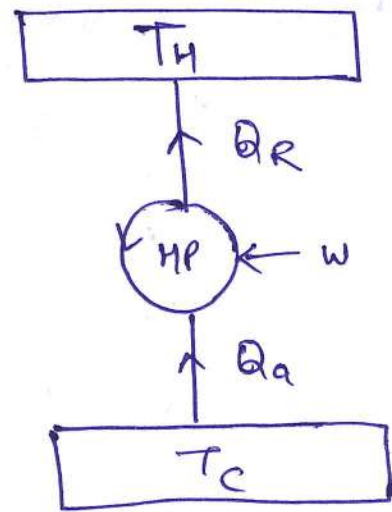
$$\boxed{(\text{COP})_{\text{HP}} = 5}$$

Also, $(\text{COP})_{\text{HP}} = \frac{Q_R}{W}$

$$\begin{aligned} Q_R &= (\text{COP})_{\text{HP}} \times W \\ &= 5 \times 1 \end{aligned}$$

$$\boxed{Q_R = 5 \text{ kW}}$$

Ans



Q3! - A refrigeration system produces 40 kg/hr of ice at -10°C from water at 35°C . Find the refrigerating effect per hour. Also, represent it in terms of tonnes of refrigeration. If it consumes 1 kW of energy to produce ice, find the COP. Take, latent heat of ice as 335 kJ/kg , specific heat of water as 4.18 kJ/kgK , and specific heat of ice as 2.108 kJ/kgK .

Solution:- Given data,

$$T_w = 35^\circ\text{C} = 308\text{K}$$

$$T_i = -10^\circ\text{C} = 263\text{K}$$

$$W = 1\text{KW}$$

$$m = 40\text{ KJ/hr}$$

$$C_w = 4.18\text{ KJ/kgK}$$

$$C_i = 2.108\text{ KJ/kgK}$$

$$L_i = 335\text{ KJ/kg}$$

(i) Amount of heat to be removed,

Refrigerating effect, $Q_a = (m c_p)_w + m L + (m c_p)_{ice}$

$$= \{40 \times 4.18 \times (35-0)\} + \{40 \times 335\} \\ + \{40 \times 2.108 \times (0+10)\}$$

$$\boxed{R.E. = 20,095.12\text{ KJ/hr}}$$

$$(ii) \quad RE = \frac{20,095.12}{3600}\text{ KJ/sec}$$

$$\boxed{R.E. = 5.582\text{ KJ/sec}}$$

Also, $1\text{TR} = 3.5\text{ KW}$

$$\Rightarrow R.E. = \frac{5.582}{3.5}\text{ TR}$$

$$\boxed{RE = 1.59\text{ TR}}$$

$$(iii) \quad (COP)_R = \frac{Q_a}{W} = \frac{5.582}{1}$$

$$(COP)_R = 5.582$$

Q4 → 1.5 kW per tonne of ~~refrigerant~~ refrigeration is ~~is~~ required to maintain the temperature of -40°C in the refrigerator. If the refrigeration cycle works on Carnot, determine the following,
 (i) COP of the cycle, (ii) Temperature of the sink
 (iii) Heat rejected to the sink per TR,
 (iv) COP if used as a heat pump.

(AKTU 2022-23 Even Sem)

Solution! $T_c = -40^\circ\text{C} = 233\text{ K}$

$W = 1.5\text{ kW per TR}$

as we know, $1\text{ TR} = 3.5\text{ kW}$

$$(i) \text{ thus, } (COP)_R = \frac{3.5}{1.5} \\ = 2.33$$

(ii) Also, for a Carnot Cycle,

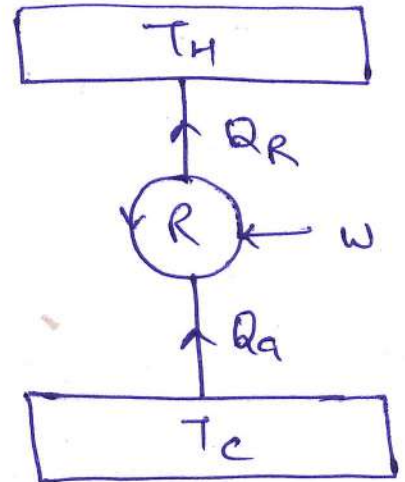
$$(COP)_R = \frac{T_c}{T_H - T_c}$$

$$2.33 = \frac{233}{T_H - (233)}$$

$$2.33 T_H - 2.33 \times 233 = 233$$

$$T_H = \frac{3.33 \times 233}{2.33}$$

$$\boxed{T_H = 333 \text{ K}} \quad \text{or} \quad \boxed{T_H = 60^\circ \text{C}}$$



(iii)

$$(COP)_{HP} = 1 + (COP)_R = 1 + 2.33$$
$$= \underline{3.33}$$

Also,

$$(COP)_{HP} = \frac{Q_R}{W}$$

$$Q_R = 3.33 \times 1.5$$
$$= 5 \text{ kW/TR}$$

(iv)

$$(COP)_{HP} = 3.33.$$

Q5 → A cold storage is to be maintained at -5°C , while the surroundings are at 35°C .

The heat leakage from the surrounding into the cold storage is estimated to be 29 kW. The actual COP of the refrigeration plant is one-third of an ideal plant working between same temperatures. Find the power required to drive the plant.

(AKTU 2023-24 Odd Sem)

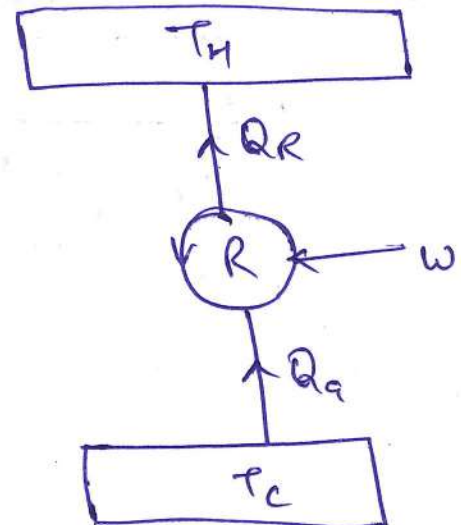
Solution! Given data

$$T_c = -5^{\circ}\text{C} = 268\text{ K}$$

$$T_h = 35^{\circ}\text{C} = 308\text{ K}$$

$$(\text{COP})_{\text{actual}} = \frac{1}{3} (\text{COP})_{\text{ideal}}$$

$$Q_a = 29\text{ kW}$$



$$\begin{aligned} \text{We know that, } \left[(\text{COP})_R \right]_{\text{ideal}} &= \frac{T_c}{T_h - T_c} \\ &= \frac{268}{308 - 268} \end{aligned}$$

$$\left[(\text{COP})_R \right]_{\text{ideal}} = 6.7$$

$$(\text{COP})_{\text{actual}} = \frac{1}{3} \times 6.7 = 2.23$$

$$\text{Also, } (\text{COP})_{\text{actual}} = \frac{Q_a}{W}$$

$$\Rightarrow W = \frac{29}{2.23}$$

$$W = 13 \text{ kW}$$

Ans

Q6 → What should be the capacity of an air-conditioning plant which has round the clock cooling load of 300 MJ/day ?

GATE 1993

Ans →

Refrigerating effect = 300 MJ/day

$$= \frac{300 \times 10^3}{24 \times 60 \times 60} \frac{\text{kJ}}{\text{sec}}$$

$$= 3.47 \text{ kJ/sec}$$

as we know that,

$$\text{ITR} = 3.5 \text{ kJ/sec}$$

$$\Rightarrow \text{R.E.} = \frac{3.47}{3.5} = 0.99$$

So, 1 ton of refrigeration Air conditioning plant will suffice.

Q73 The value of enthalpy at the beginning of compression, at the end of compression, at the end of condensation are 185 kJ/kg , 210 kJ/kg , and 85 kJ/kg respectively. What is the value of COP of vapour compression refrigeration system?

IES-2005

Solution:- Given data

$$h_1 = 185 \text{ kJ/kg}$$

$$h_2 = 210 \text{ kJ/kg}$$

$$h_3 = 85 \text{ kJ/kg}$$

but, since, $h_3 = h_4$

$$\Rightarrow h_4 = 85 \text{ kJ/kg}$$

as we know, for a VCRS,

$$(COP)_R = \frac{h_1 - h_4}{h_2 - h_1} = \frac{185 - 85}{210 - 185}$$

$$(COP)_R = 4$$

